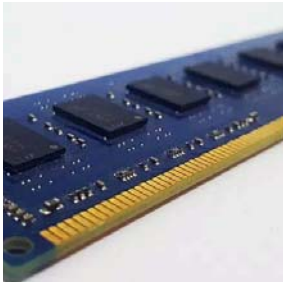


## A 1.1.4 Primary Memory



### A 1.1.4 Explain the purposes of different types of primary memory

When the CPU needs an instruction or a value, where that data happens to be decides how long the CPU waits for it. Primary memory is not one thing but a family of types, each with a distinct purpose: RAM is the working memory, ROM holds the permanent startup instructions, cache is the high-speed staging area between the CPU and RAM, and registers are the CPU's own workbench. This page explains the purpose of each type, how the CPU works through them in order to keep itself busy, and what the terms cache hit and cache miss mean for performance.

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### Key Questions

- What purpose does each type of primary memory serve, and why is each designed the way it is?
- In what order does the CPU look for data, and why does that order optimise performance?
- What are a cache hit and a cache miss, and why does the hit rate matter?

### IB Command Term note

**EXPLAIN:** Give a detailed account including reasons or causes. Naming what each memory type stores is description; explaining why it is built that way (why RAM is volatile, why ROM is read-only, why cache exists at all) is what this statement assesses.

**WHAT TO INCLUDE:** RAM, ROM, cache (L1, L2, L3) and registers; the order in which the CPU works through them and why; and the meaning and relevance of cache hit and cache miss.

## The four types and their purposes

**RAM (random-access memory)** is the computer's working memory. It holds the operating system, the programs currently running, and the data the CPU is using right now. RAM is volatile: switch off the power and its contents vanish. That is acceptable because everything in RAM is a working copy; the permanent versions live in storage.

**ROM (read-only memory)** is the opposite to RAM. It is non-volatile, retaining its contents without power, because its job is to hold the instructions the computer needs before anything else is available: the firmware, including the BIOS that boots the machine. It is read-only for the same reason it is permanent: startup instructions must survive power-off and must not be corrupted by a running program.

**Cache** is a small amount of very fast memory that sits between the CPU and RAM, holding the data and instructions used most often. It exists because RAM, fast as it is, still keeps the CPU waiting. Cache comes in levels: L1 is the smallest and fastest, built into each core; L2 is larger and slightly slower; L3 is the largest and slowest of the three, shared by all the cores.

**Registers** are the fastest memory of all: tiny storage locations inside the CPU holding exactly what it is working on this instant, such as operands, addresses and intermediate results. A 1.1.1 named the five you must know individually (PC, IR, MAR, MDR, AC).

## How the CPU works with memory

### THINK FIRST

If registers are the fastest memory, why not build the whole computer out of them? Before reading on, consider what the catch must be, given that real machines give you gigabytes of RAM but only a few kilobytes of cache per core.

The catch is cost and physical size: the faster the memory, the more expensive it is per byte and the less of it can sit close to the CPU. The hierarchy is how these memories are ordered.

The CPU looks for what it needs in speed order: registers first, then L1, L2 and L3 cache, and only then RAM. Each step down is larger but slower. On a typical modern processor a register or L1 access takes a few cycles, while a trip to RAM can cost hundreds.

The key principle is to keep the data needed soonest in the fastest level, so the slow trips happen as rarely as possible. Beyond primary memory sits secondary storage, slower again by a large factor. This is covered later in this topic.

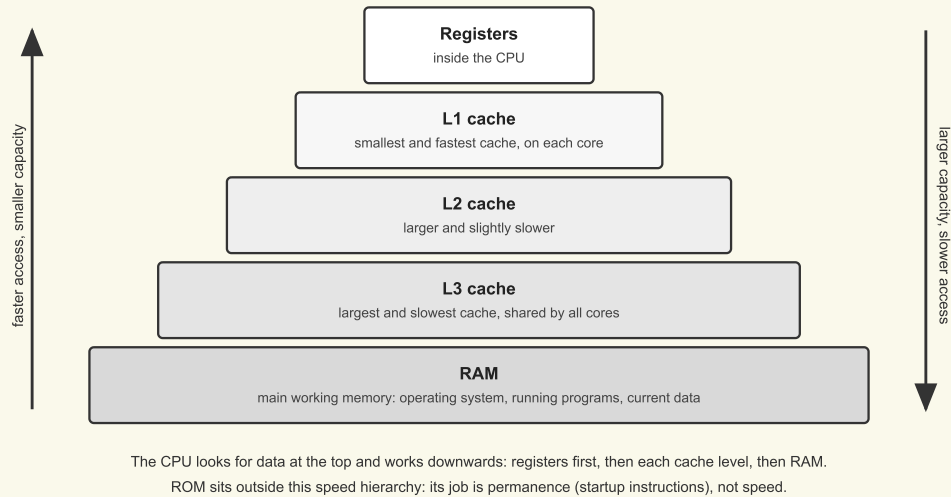


Figure 1. The primary memory hierarchy: the CPU searches from registers down to RAM, trading capacity for speed at every level.

## Cache hits and cache misses

**A cache hit** is the good outcome: the CPU finds what it needs in cache and proceeds almost immediately.

**A cache miss** is the slower path: the data is not in cache, so the CPU must fetch it from RAM, which takes much longer. The fetched data is normally copied into cache as it arrives, so the next request for it will hit. A miss is not an error and nothing has gone wrong; it is simply the price paid the first time data is needed, or when the cache has had to remove it to make room.

This is why the hit rate, the proportion of accesses that are hits, makes a big difference. A program whose data fits comfortably in cache runs dramatically faster than one that misses constantly, even on identical hardware, because every miss stalls the CPU for hundreds of cycles while RAM responds.

### COMMON EXAM MISTAKE

Writing that a cache miss means an error, a crash, or data being lost. A miss is a normal, frequent event: the CPU simply continues its search in RAM, and usually copies the data into cache so the next access hits. The cost of a miss is time, nothing else.

# Summary

## SUMMARY REFERENCE

| TYPE               | VOLATILE? | PURPOSE                                                                                                                                      |
|--------------------|-----------|----------------------------------------------------------------------------------------------------------------------------------------------|
| Registers          | Yes       | Hold what the CPU is working on this instant: operands, addresses, intermediate results. Fastest of all.                                     |
| Cache (L1, L2, L3) | Yes       | Keep frequently used data and instructions close to the CPU. L1 smallest and fastest, per core; L2 larger; L3 largest, shared by all cores.  |
| RAM                | Yes       | Working memory: the operating system, running programs and current data. Volatility is the price of speed; permanent copies live in storage. |
| ROM                | No        | Permanent startup instructions: firmware and the BIOS. Read-only so they survive power-off and cannot be corrupted.                          |

## Try it on paper

**Explain why ROM rather than RAM is used to store the instructions a computer needs at startup. [2 marks]**

**Model answer:** RAM is volatile, so anything stored in it is lost when the power is off; the startup instructions must be available the moment the machine is switched on, before anything has been loaded. [1] ROM is non-volatile and read-only, so the instructions survive without power and cannot be overwritten or corrupted by running software. [1]

**Explain the difference between a cache hit and a cache miss, and why a higher hit rate improves performance. [3 marks]**

**Model answer:** A cache hit occurs when the CPU finds the data it needs in cache and can proceed almost immediately. [1] A cache miss occurs when the data is not in cache, forcing the CPU to fetch it from RAM, which takes many times longer. [1] A higher hit rate means fewer of these slow RAM accesses, so the CPU spends less time stalled waiting for data and the program runs faster. [1]

## A 1.1.4 Quiz

1

Which type of primary memory is volatile, meaning it loses its contents when the power is turned off?

A. ☐ Registers

B. ☐ Cache

C. ☒ RAM

D. ☐ ROM

2

What is the primary purpose of ROM in a computer system?

A. ☐ To store user data and documents.

B. ☒ To store the BIOS and firmware, which are essential for startup.

C. ☐ To store frequently accessed data for faster processing.

D. ☐ To store the operating system and running applications.

3

Which level of cache memory is the smallest and fastest, located closest to the CPU?

A. ☐ All cache levels have the same speed and size.

B. ☐ L2 cache

C. ☒ L1 cache

D. ☐ L3 cache

4

What happens when the CPU needs to access data that is not stored in any of the cache levels?

- A. ☐ The CPU accesses the hard drive.
- B. ☐ The program crashes.
- C. ☐ The CPU waits for the data to be loaded into the cache.
- D. ☒ The CPU accesses the RAM

5

Which statement BEST describes a cache hit?

- A. ☐ The CPU fails to find the required data in the cache memory.
- B. ☐ The CPU clears the cache memory.
- C. ☒ The CPU finds the required data in the cache memory.
- D. ☐ The CPU writes data to the cache memory.

6

What is the main advantage of having a higher cache hit rate?

- A. ☐ It increases the storage capacity of the RAM
- B. ☐ It allows the computer to run more applications simultaneously.
- C. ☒ It improves the overall performance of the computer by speeding up memory access.
- D. ☐ It reduces the cost of the computer.

7

Which of the following is NOT stored in registers?

- A. ☒ The entire operating system.
- B. ☐ The address of the next instruction.
- C. ☐ Operands for calculations.
- D. ☐ Intermediate results of calculations.

8

How does the CPU interact with different memory types to optimise performance?

- A. ☒ It accesses memory types in a hierarchical manner, starting with the fastest
- B. ☐ It accesses all memory types simultaneously.
- C. ☐ It only uses cache memory, ignoring RAM and registers.
- D. ☐ It randomly selects a memory type to access.

9

What is a common misconception about cache memory?

- A. ☐ Cache memory is slower than RAM.
- B. ☐ Cache memory is non-volatile.
- C. ☒ Cache memory is larger than RAM.
- D. ☐ Cache memory stores all the data that the CPU needs.

10

Why is understanding the concept of cache misses important?

- A. ☒ It helps in identifying and resolving performance bottlenecks related to memory access.
- B. ☐ It helps in increasing the storage capacity of the hard drive.
- C. ☐ It helps in reducing the power consumption of the CPU.
- D. ☐ It helps in choosing the right type of RAM for a computer.

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